

Problem 9

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Problem 1 Compute the integral: $\int \sec^2(x) \tan(x) \, dx$.

Explanation. Let $u = \tan(x)$. Then $du = \sec^2(x)dx$, and so

$$\begin{aligned} \int \sec^2(x) \tan(x) \, dx &= \int u \, du \\ &= \frac{1}{2} u^2 + C \\ &= \frac{1}{2} \tan^2(x) + C. \end{aligned}$$

NOTE: The substitution $v = \sec(x)$ would also work to solve this problem, as would the substitution $w = \sec^2(x)$. It is a good exercise to work these out!